

SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA11

COURSE OUTCOME COVERED

CO3: Analyze system requirements using object-oriented techniques to identify classes, attributes, and relationships and develop use case models. (BL4)

Assessment Tool Weightage: 40%

QUESTIONS:16

Total Marks:50

Annexure A

Experiments

Code of Conduct:

I Annajothi S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Annajothi S

RUBRICS FOR CALCULATION:

Experiments

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

16 ee) Develop a system using UML for implementing the Book Bank process. It should verify the details of the student by the central computer. The details regarding the student will be provided to the central computer through the administrator in the book bank and the computer will verify the details of the student and provide approval to the office. Then the books that are needed by the student will issue from the office to him.

UML Design for Book Bank System (Detailed Explanation)

Introduction

A **Book Bank System** is used in educational institutions to lend books to eligible students for a semester or academic year. The system ensures that only authorized students receive books after their details are verified by a central computer system.

In this scenario:

- The **student** requests books from the Book Bank.
- The **administrator** collects and enters the student's details into the system.
- The **central computer** verifies the student's eligibility.
- After successful verification, an **approval** is sent to the **Book Bank Office**.
- The **Book Bank Office** issues the required books to the student and updates its records.

Requirements of the System

Functional Requirements

- Register student details.
- Verify student information through the central computer.
- Approve or reject the student's request.
- Maintain book inventory.
- Issue books to eligible students.
- Record issued books and update availability.

Non-Functional Requirements

- Secure handling of student information.
- Fast verification process.
- Accurate record maintenance.
- Reliable communication between administrator, central computer, and office.

Actors Involved

Actor	Role
Student	Requests books and provides personal details.
Administrator	Enters student information into the system and forwards it for verification.

Actor

Role

Central Computer

Verifies eligibility and approves or rejects the request.

Book Bank Office Issues books after receiving approval and updates records.

Use Case Diagram Explanation

Main Use Cases

1. Provide Student Details

- The student submits information such as roll number, name, department, and year.

2. Submit Details

- The administrator enters the student's information into the system.

3. Verify Student Details

- The central computer checks whether the student satisfies the eligibility criteria.

4. Approve Request

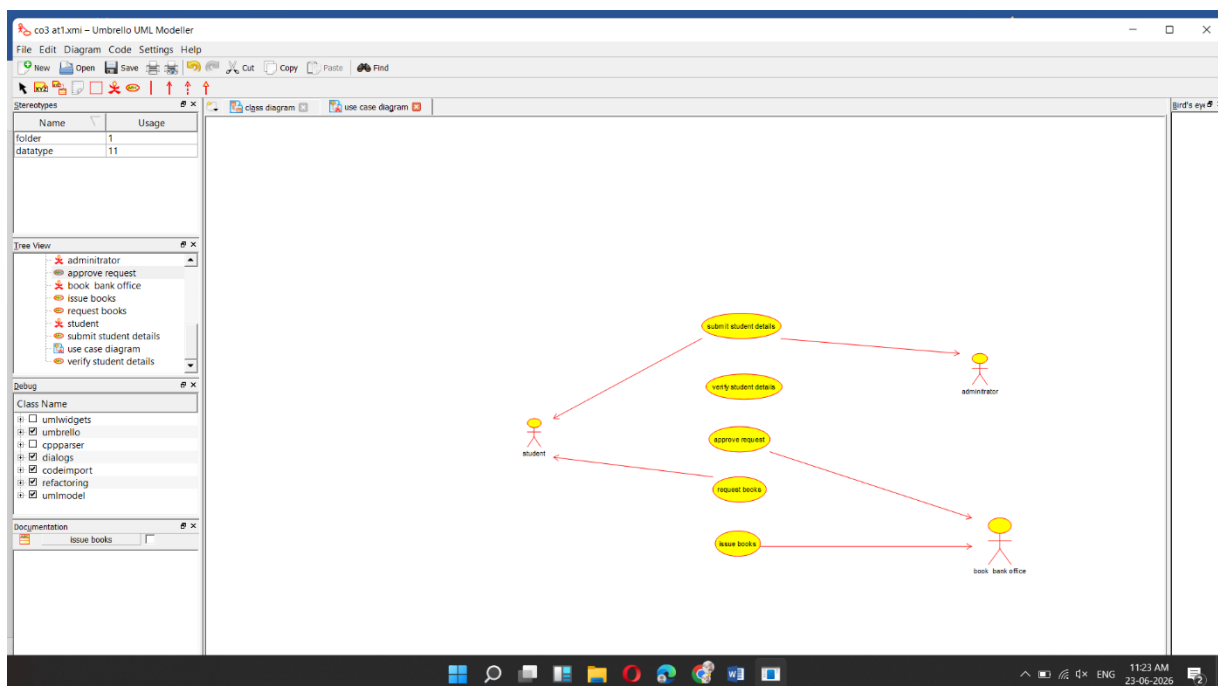
- If verification succeeds, approval is generated.

5. Request Books

- The student requests one or more books.

6. Issue Books

- The office issues available books to the approved student and records the transaction.



Class Diagram Explanation

The Book Bank System can be represented using the following classes.

Student Class

Attributes

- Student ID
- Name
- Department
- Year
- Eligibility Status

Methods

- Request Book
- View Status

Administrator Class

Attributes

- Admin ID
- Name

Methods

- Enter Student Details
- Send Details for Verification

CentralComputer Class

Attributes

- Student Database
- Verification Rules

Methods

- Verify Student
- Approve Request
- Reject Request

BookBankOffice Class

Attributes

- Office ID
- Issue Register

Methods

- Issue Book
- Receive Approval
- Update Inventory

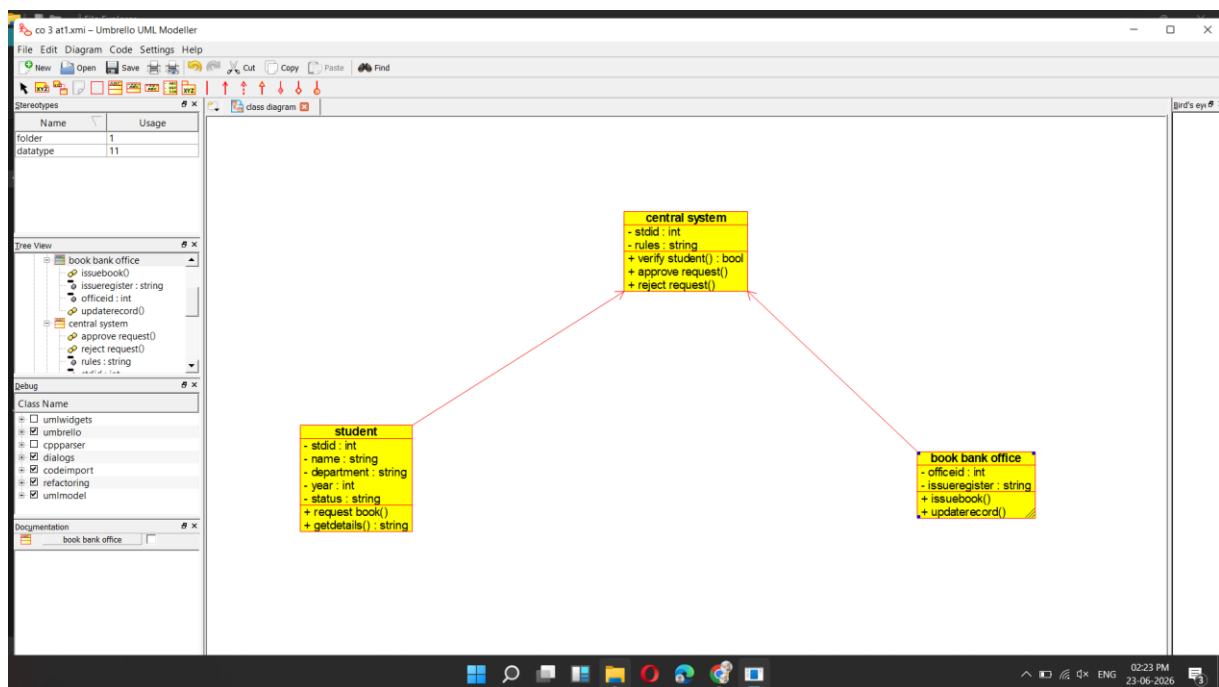
Book Class

Attributes

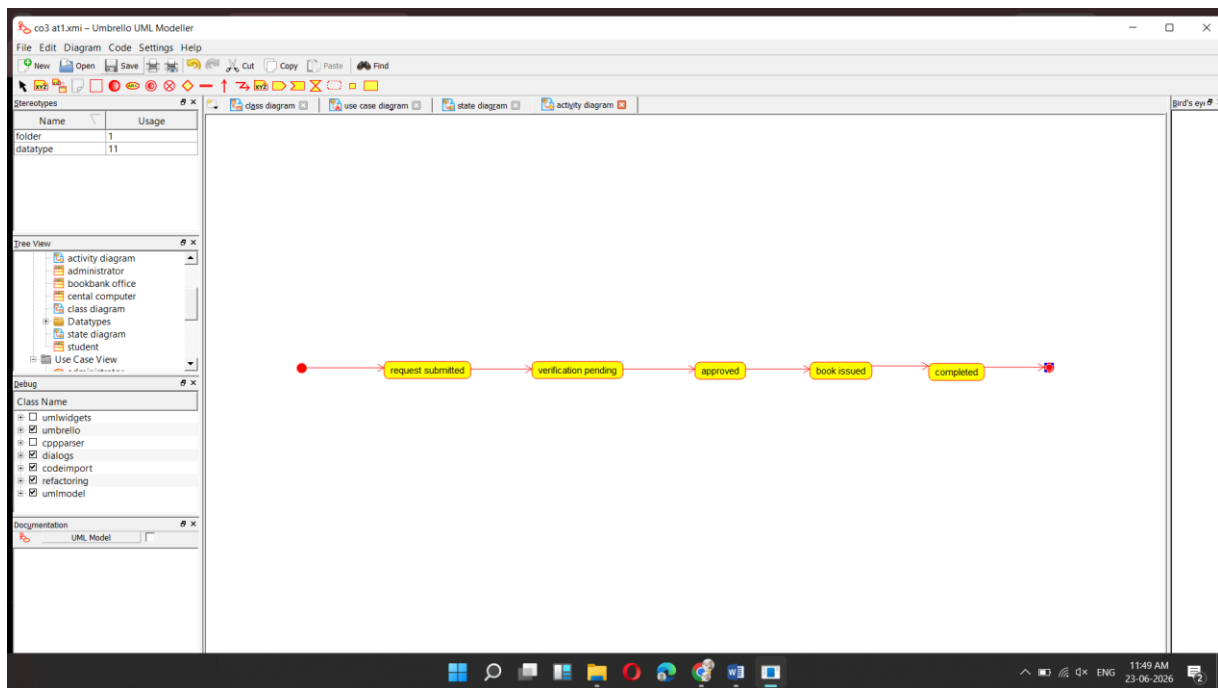
- Book ID
- Title
- Author
- Availability

Methods

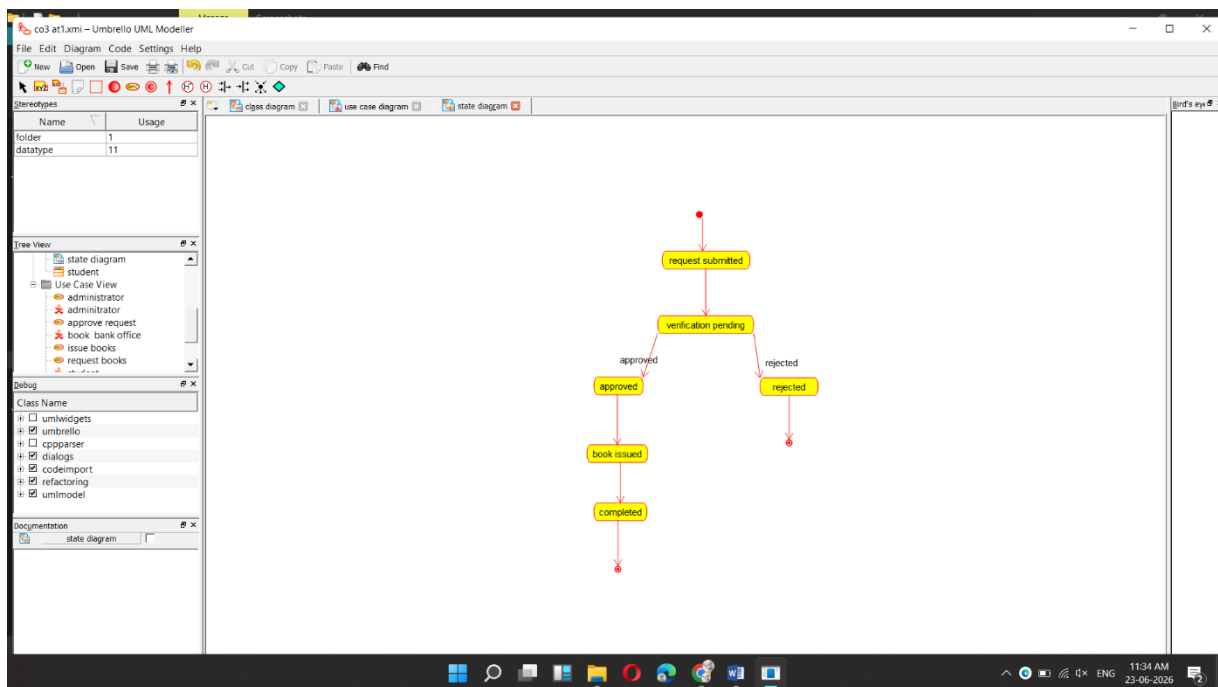
- Check Availability
- Update Status



State Diagram:



Activity Diagram:

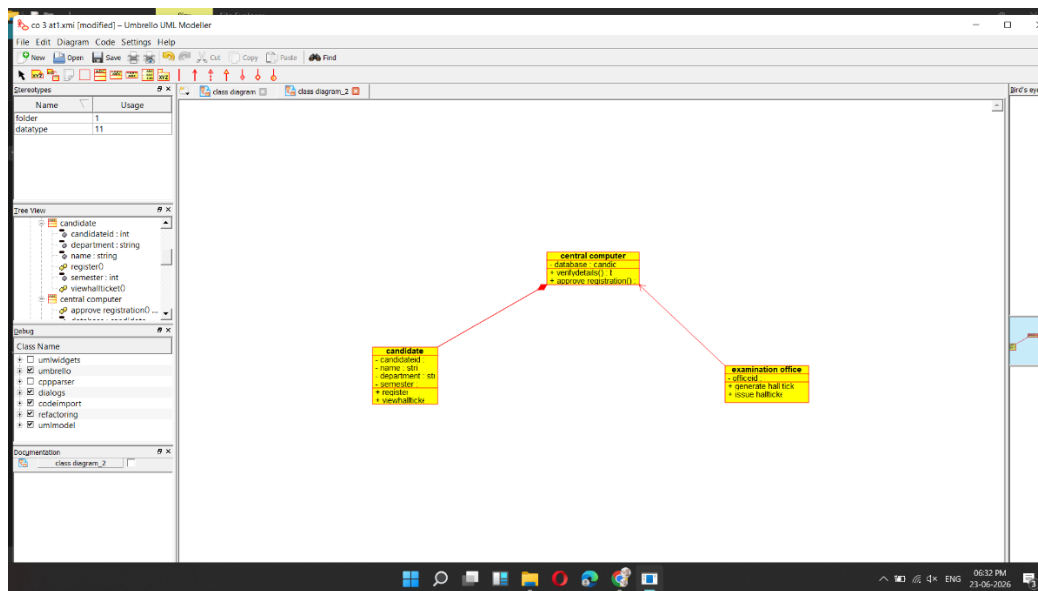


ff) Develop a system using UML for implementing the Exam Registration Portal. It should verify the details of the candidate by the central computer. The details regarding the candidate will be provided to the central computer through the administrator and the computer will verify the details of the candidate and provide approval. Then the hall ticket will be issued from the office to the candidate.

Class Diagram:

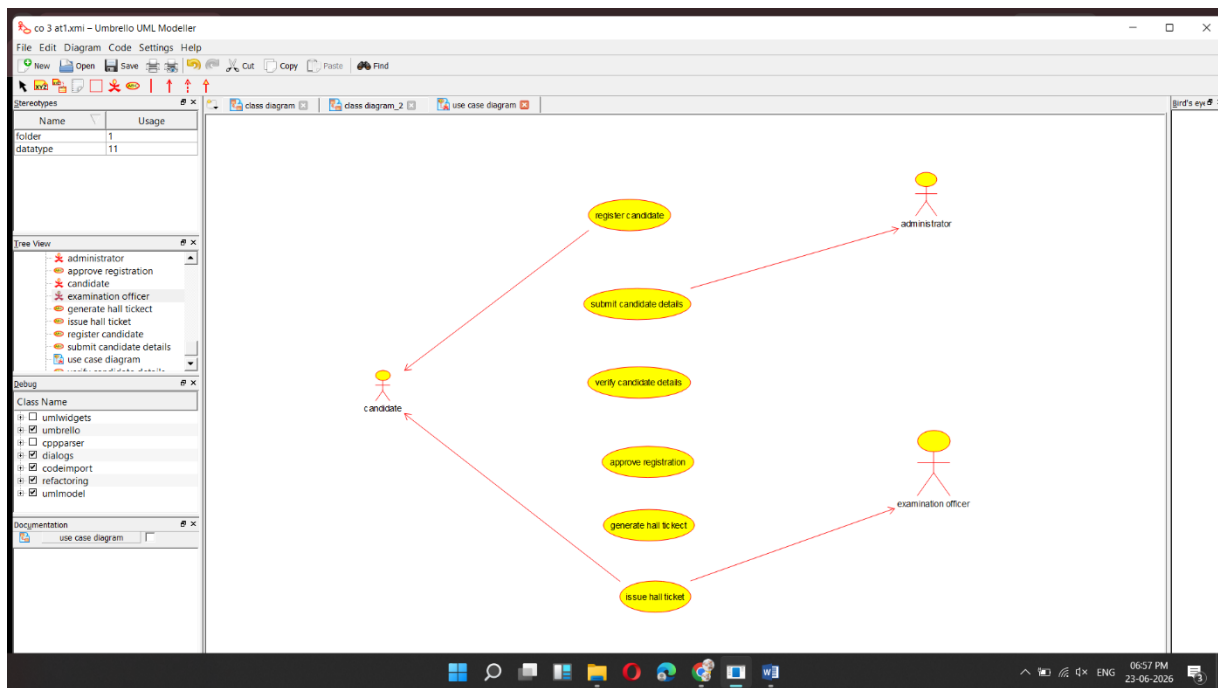
The **Class Diagram** represents the static structure of the system using three classes:

- **Candidate:** Stores candidate information and provides functions such as registration and viewing the hall ticket.
 - **Central Computer:** Verifies candidate details and approves the registration.
 - **Examination Office:** Generates and issues the hall ticket after approval.
- The associations between these classes show how candidate information flows through verification to hall ticket issuance.



Use case diagram:

The **Use Case Diagram** shows the interactions between the external users and the **Exam Registration Portal**. The **Candidate** registers and requests a hall ticket, the **Administrator** submits the candidate's details for verification, and the system verifies the information, approves the registration, generates the hall ticket, and finally the **Examination Office** issues it to the candidate.



State diagram:

The State Diagram shows the different states of an exam registration request throughout its lifecycle. The request moves from Registration Submitted to Verification Pending, then to Approved, followed by Hall Ticket Generated, Hall Ticket Issued, and finally Completed. It illustrates how the registration changes state until the process is successfully finished.

